Dr. Ebuka Philip Oguchi Ph.D.

Contact Information

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Academic Appointment

Assistant Professor (Tenure Track)
Department of Computer and Information Sciences
Spelman College
January 2026 – Present

Research Interests

Cybersecurity; authentication and message integrity verification; trust establishment for emerging wireless systems; physical-layer security; wireless and mobile security; Agricultural Internet of Things; autonomous and vehicular network security; trustworthy AI; distributed confidential computing; RF fingerprinting; location verification; molecular and nano-scale communication security; applied machine learning for adversarial environments; cyber-physical systems.

Education

University of Nebraska–Lincoln
Ph.D. in Computer Science, August 2025
Advisor: Prof. Nirnimesh Ghose
Dissertation:
Authentication and Message Integrity Verification for Emerging Wireless Networks
GPA: 4.0/4.0

Changchun University of Science and Technology, China
M.S. in Computer Applied Technology, 2020
Advisor: Prof. Fang Ming
Thesis:
Smoke Recognition Using ResNet and GoogleNet

Nnamdi Azikiwe University, Nigeria
B.Eng. in Electronic and Computer Engineering, 2016
Senior Project:
Biometric Course Registration System Using Fingerprint Authentication

Research Support

- **Senior Personnel/ Collaborator**, NSF Center for Distributed Confidential Computing (CDCC) 2026–Present
  Multi-institutional NSF Frontiers Center supporting research in confidential computing, secure distributed systems, and trustworthy computing.

- **Collaborator**, Google.org Cybersecurity Clinic 2026–Present
  Supporting the establishment of the Spelman College Cybersecurity Clinic and integration of clinic activities into the undergraduate Cybersecurity curriculum.
- **Graduate Research Support**
  Supported through NSF awards CNS-2331191, CNS-2225161, CNS-1619285, and ECCS-2030272.

Publications

Peer-Reviewed Journal Articles

1. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
   Soil-Assisted Trust Establishment for Underground Wireless Networks.
   IEEE Transactions on Dependable and Secure Computing (TDSC).
   Accepted for Publication, 2026.
2. Anderson, M.*; Duong, T.T.*; **Oguchi, E.***; Wisniewska, A.; Ghose, N.
   A Review of Confidentiality, Integrity, and Availability Challenges in Molecular Communications.
   IEEE Transactions on Molecular, Biological, and Multi-Scale Communications.
   Accepted for Publication, 2026.

   *Equal contribution.

Peer-Reviewed Conference Papers

1. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
   VET: Autonomous Vehicular Credential Verification Using Trajectory and Motion Vectors.
   EAI SecureComm, Hong Kong, 2023.
2. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
   STUN: Secret-Free Trust Establishment for Underground Wireless Networks.
   IEEE INFOCOM Workshops, 2022.

Manuscripts Under Review

1. **Oguchi, E.**; Lado, H.; Evans, R.; Menon, A.; Wang, B.; Ghose, N.
   LocaRadio: RF Fingerprint-Based Location Authentication for Wireless Networks.
   Submitted to IEEE CCNC.

Dissertation and Theses

1. **Oguchi, E.**
   Authentication and Message Integrity Verification for Emerging Wireless Networks.
   Ph.D. Dissertation, University of Nebraska–Lincoln, 2025.
2. **Oguchi, E.**
   Smoke Recognition Using ResNet and GoogleNet.
   M.S. Thesis, Changchun University of Science and Technology, 2020.

3. **Oguchi, E.**
Biometric Course Registration System Using Fingerprint Authentication.
B.Eng. Project, Nnamdi Azikiwe University, 2016.

Research Experience

Research Assistant 2021–2025
Department of Computer Science and Engineering
University of Nebraska–Lincoln

- Developed the **STUN (Secret-Free Trust Establishment for Underground Wireless Networks)** protocol, a physical-layer authentication framework for secure bootstrapping in Agricultural Internet of Things (Ag-IoT) environments. The work was published at IEEE INFOCOM Workshops and later extended into an accepted journal article in IEEE Transactions on Dependable and Secure Computing.
- Developed the **VET (Vehicular Credential Verification)** framework for autonomous and connected vehicle networks by integrating trajectory information, motion vectors, Doppler characteristics, and wireless signal measurements to verify transmitter authenticity.
- Investigated RF fingerprinting and location authentication techniques using wireless channel characteristics, signal processing, and machine learning for underground and over-the-air communication environments.
- Conducted research on lightweight security mechanisms for molecular and nano-scale communication networks, including authentication, message integrity, and threat modeling for biological communication systems.
- Designed and implemented wireless experimental testbeds using Software Defined Radios (USRPs), GNU Radio, MATLAB, Python, and embedded hardware platforms.
- Contributed to externally funded NSF research projects involving secure wireless systems, underground communications, cyber-physical systems, and confidential distributed computing.

Teaching Experience

Assistant Professor (Tenure Track) Jan. 2026–Present
Department of Computer and Information Sciences
Spelman College

- SCIS 216: Computer Organization and Design
- SCIS 216L: Computer Organization and Design Laboratory
- SCIS 123: C++ Programming
- Developed laboratory exercises involving digital logic, assembly language programming, breadboard implementation, and computer architecture.
- Mentored undergraduate students through course projects, independent studies, and research activities.

Teaching Assistant Fall 2022
Cryptography and Security (CSCE 477/877)
University of Nebraska–Lincoln

- Assisted with laboratories covering RSA, AES, hash functions, digital signatures, authentication protocols, and network security.

- Guided students in implementing cryptographic algorithms and analyzing protocol security.

Teaching Assistant Jan. 2018–May 2018
Computer Architecture and Software Laboratory
D.S. Adegbenro ICT Polytechnic, Nigeria

Undergraduate Research Mentoring

- Mentor undergraduate students through independent studies and faculty-directed research projects.
- Supervise projects involving wireless security, cybersecurity, Software Defined Radios, embedded systems, machine learning, and trustworthy AI.
- Mentor students through literature reviews, software implementation, experimental design, data analysis, manuscript preparation, and conference presentations.
- Support students pursuing Honors theses and undergraduate research experiences.
- Prepare students for graduate school through research advising, fellowship applications, technical writing, and presentation skills.
- Encourage participation in undergraduate research symposia, conference poster sessions, and peer-reviewed publications whenever appropriate.

Professional Service

- Reviewer, IEEE Transactions on Dependable and Secure Computing (2026–Present)
- Reviewer, IEEE Transactions on Industrial Informatics
- Reviewer, ACM Transactions on Cyber-Physical Systems
- Reviewer, ACM WiSec
- Reviewer, EAI SecureComm
- Judge, Spelman College Hack-A-Bot Competition (2026)
- Judge, University of Nebraska Engineering Poster Competition
- Member, Departmental Curriculum Review Committee, Spelman College

Professional Development

- Fellow, ACS Teaching and Learning Workshop, Rollins College (2026)
- Faculty Onboarding and Mentorship Program, Spelman College (2026–Present)
- CITI Responsible Conduct of Research Training

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- National Society of Black Engineers (NSBE)
- Google Computer Science Research Mentorship Program (CSRMP)
- Institute for African-American Mentoring in Computing Sciences (iAAMCS)

Awards and Honors

- Outstanding Ph.D. Dissertation Award, University of Nebraska–Lincoln (2025)
- Mary E. and Elmer H. Dohrmann Fellowship, University of Nebraska–Lincoln (2023)
- Chinese Government Scholarship (2018)
- Outstanding Graduate Student Recognition, University of Nebraska–Lincoln

Invited Activities

- Senior Personnel, NSF Center for Distributed Confidential Computing (CDCC)
- Collaborator, Google.org Cybersecurity Clinic Initiative
- Invited Judge, Spelman College Hack-A-Bot Competition
- Poster Judge, University of Nebraska College of Engineering Research Fair

Technical Skills

Programming: Python, C, C++, MATLAB, Java, Assembly (MIPS)

Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn

Wireless Systems: Software Defined Radio (USRP), GNU Radio, RF Fingerprinting, Wireless Testbeds

Cybersecurity: Cryptography, Authentication, Message Integrity, Network Security, Physical-Layer Security, Penetration Testing, NIST Cybersecurity Framework

Hardware: Embedded Systems, Breadboard Prototyping, Electronic Circuit Design, Microcontrollers

Research Tools: Linux, Git, LaTeX, Microsoft Azure, Docker